# 🧩 Detailed BRD: Inventory Management

### Overview

The Inventory Monitoring module is the central command center for the hardware store owner (Admin). It provides real-time visibility into the current stock levels of all products, tools, and materials. Beyond simple tracking, this module actively works to prevent stockouts by analyzing sales velocity to generate automated restocking suggestions. It also provides the necessary tools to manage the product catalog, update pricing, and make manual adjustments for lost or damaged goods.

### Scope

- Viewing a comprehensive master list of all hardware items and current stock levels
- Adding new products to the system catalog (defining name, ID, unit price, and initial stock)
- Editing existing product details or deactivating discontinued items
- Manual stock adjustments (adding restocked deliveries or deducting damaged/lost items)
- Automated low-stock visual indicators
- Demand forecasting and automated order suggestions based on recent sales trends

### Out-of-Scope

- Automated Purchase Order (PO) generation sent directly to external suppliers
- Multi-warehouse or multi-branch inventory syncing
- Supplier database management or price comparison tools
- Barcode generation and printing

### Functional Requirements

| Requirement | Actor | Priority | Rationale |
|---|---|---|---|
| The system shall display a master inventory list detailing item name, current stock quantity, and unit price. | Admin | Must | Core requirement for monitoring what is physically available in the store. |

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| The system shall allow the user to add new hardware items to the database, specifying a unique ID, name, price, and starting quantity. | Admin | Must | Allows the store to expand its catalog as they source new products. |
| The system shall allow the user to manually adjust the stock quantity of an existing item (e.g., logging new deliveries or deducting damaged goods). | Admin | Must | Ensures the digital system can be corrected to match physical stock audits. |
| The system shall display a distinct visual alert (e.g., a red highlight) for any item whose stock drops below a predefined threshold (e.g., < 10 units). | System | Must | Provide immediate warning for critically low items before they run out. |

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| The system shall generate "Order Suggestions" detailing which items to buy and in what quantities, calculated by analyzing the demand for that item. | System | Must | Satisfies the client's core requirement to automate demand forecasting. |
| The system shall allow the user to update the selling price of existing items. | Admin | Should | Prices fluctuate, and the Admin needs control over the catalog pricing. |
| The system shall restrict all access to this module; it must be completely inaccessible to accounts with the Staff role. | System | Must | Enforces the strict 2-tier role security policy. |

**Non-Functional Requirements**

| Category | Requirement |
|---|---|
| Security | Direct URL navigation to the inventory dashboard by an unauthenticated or Staff-level user shall result in an immediate redirect to the login or POS screen. |

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| Performance | The master inventory list must load and render within 3.0 seconds, even if the database contains over 1,000 distinct hardware items. |
| Usability | The master list must include a persistent search bar and column sorting (e.g., sort by lowest stock first) without requiring page reloads. |
| Reliability | The Demand Forecasting algorithm shall run as a nightly scheduled task (or background process) so that loading the suggestions page does not slow down the application. |

**Constraints**

- The Demand Forecasting formula relies strictly on historical sales data recorded within SMART-STOCK; it cannot pull external market trends.
- No automatic supplier purchasing; the Admin must physically place the orders based on the system's suggestions.

**Limitations**

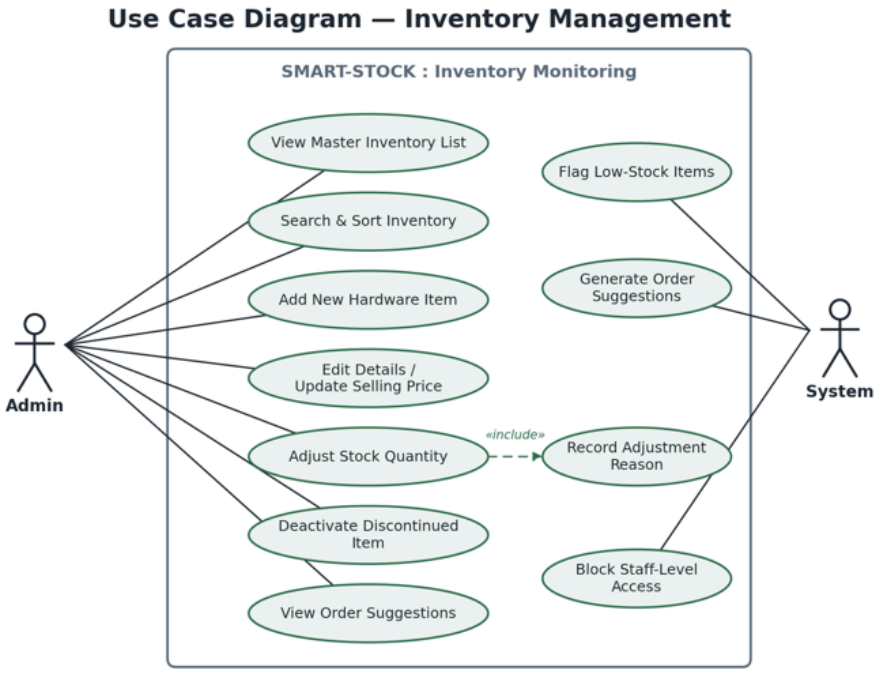
- The accuracy of the inventory entirely depends on the staff’s diligence in ringing up items correctly at the POS and the Admin consistently logging new deliveries.

**Assumptions**

- The store owner (Admin) will perform periodic physical inventory counts to ensure the physical stock matches the system's digital records.
- The business operates on a standard First-In-First-Out (FIFO) retail model, and individual product expiration dates (like paint) do not need to be rigorously tracked by the system.

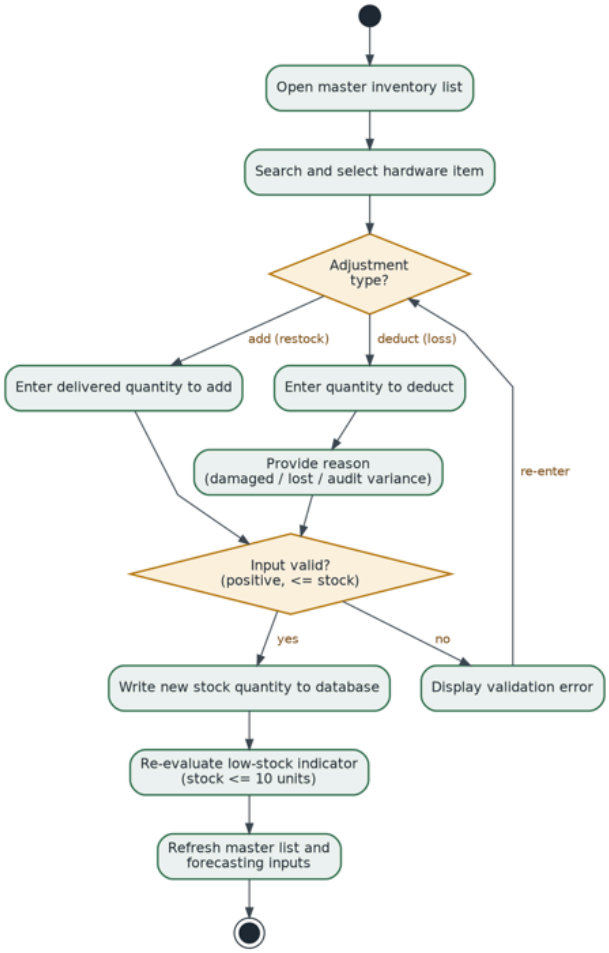
**Requirements Model**

- **Use Case Diagram**




- **Business rules**
  - Low Stock Threshold: Stock Quantity <= 10.
  - Demand Suggestion Trigger: If (Sales Demand over 30 days) > (Current Stock), add to Suggested Order List.
  - Prices must be positive numerical values.
- **Activity Diagram**
  - *Manual Stock Adjustment Workflow*

**Activity Diagram — Manual Stock Adjustment (Inventory Management)**




- **State model**
  - *Item Stock Lifecycle*

## State Model — Item Stock Lifecycle (Inventory Management)

